

M. SARANYA

Data Analyst | Business Analyst

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Data Analyst with hands-on experience in business intelligence, predictive analytics, and data visualization through internships and academic projects. Expert in Python, SQL, and Power BI, having developed 5+ predictive models and interactive dashboards. Successfully translated complex stakeholder requirements into technical deliverables, driving evidence-based decision-making across cross-functional environments.

EDUCATION

M.Sc. in Data Science | Christ University, Bengaluru | April 2026 | First Class | CGPA: 8.77/10

B.Sc. in Mathematics | Government Arts College, Karur | April 2024 | First Class Distinction | 75.71%

TECHNICAL SKILLS

Programming & Analysis: Python (Pandas, NumPy, Scikit-learn), SQL/MySQL (Advanced), Microsoft Excel (Advanced), Power BI (DAX, RLS, Data Modeling)

AI & Predictive Analytics: Linear Regression, Scikit-learn, TensorFlow, PyTorch, RAG, Sentence Transformers, EfficientNet-B0

Backend & API Development: Flask, REST APIs, JWT Authentication, Postman, GitHub

Business & Communication: Requirements Gathering, Stakeholder Communication, Financial Data Analysis, Report Writing

Statistics & Research: Hypothesis Testing, Bayesian Inference, Linear Algebra, Statistical Modelling

Cloud: AWS Certified Cloud Practitioner

INTERNSHIP EXPERIENCE

Data Analyst Intern | NMIT Solutions, Bengaluru | Feb 2026 – Aug 2026 (Ongoing)

- Engage with fintech clients through site visits to gather and document business requirements, translating stakeholder needs into analytical and technical deliverables while gaining hands-on domain exposure in financial services.
- Built a Payment Gateway Simulation System (Flask, MySQL, JWT) replicating core Razorpay workflows: merchant onboarding, OTP-based card payment verification, transaction ledger management, and webhook event notifications for payment outcomes.
- Developed DevTrack, a full-stack ticket management system supporting Admin, Developer, and Client roles with complete issue lifecycle tracking, comment threads, and stakeholder-facing reporting dashboards.
- Designed a security architecture including password hashing, role-based access control (RBAC), card number validation using the Luhn Algorithm, and protected API middleware; validated all workflows end-to-end using Postman.

Data Science Intern | Venus Solutions | Apr 2025 – May 2025

- Developed a House Price Prediction model (Python, Scikit-learn) covering the full analytical pipeline: data cleaning, feature engineering, model training, and performance evaluation using regression metrics including RMSE and R-squared.
- Deployed the prediction model as a Flask web application, enabling users to input property details through a clean UI and receive real-time price predictions based on trained model output.
- Analyzed financial datasets for local loan systems, identifying credit risk patterns and lending trends to support strategy optimization and data-driven business decision-making.
- Built interactive data dashboards (Flask) to visualize financial KPIs; queried and processed education-sector datasets using MySQL and Microsoft Excel, delivering structured analytical reports for management review.
- Communicated data findings to non-technical stakeholders through clear visualizations and presentations, translating complex financial metrics into actionable business conclusions.

Data Science Intern | Prodigy | Oct 2024 – Nov 2024

- Performed comprehensive EDA, data cleaning, and preprocessing on real-world datasets using Python (Pandas, NumPy), resolving missing values, outliers, and structural inconsistencies to improve data quality and analysis readiness.
- Developed a predictive Linear Regression model with feature selection, model training, and performance evaluation, producing statistical insights on variable relationships and model fit.
- Generated data profiling reports and visualized statistical distributions and trends, delivering structured documentation with findings and business-oriented recommendations.

PROJECTS

Quick-Commerce Operational & Customer Insights Dashboard | *Power BI · DAX · Data Modeling*

- Designed and developed a comprehensive end-to-end analytics solution covering 17 business KPIs across sales performance, customer purchasing behavior, order fulfillment efficiency, delivery operations, inventory availability, and city-level revenue analysis.
- Established a relational data model across 4 interconnected tables (Orders, Products, Delivery, Inventory), authored 15+ DAX measures (CALCULATE, DISTINCTCOUNT, DIVIDE, COUNTROWS) in a dedicated Measures Table, and created interactive visuals including KPI cards, treemap, gauge chart, donut chart, line chart, and slicers.
- Addressed critical business questions on top reordered products, on-time delivery rate, out-of-stock risk, quarterly customer order frequency trends, category demand patterns, and city-wise revenue contribution.

Secure AI-Based Banking Chatbot | *Flask · MySQL · JWT · RAG · Sentence Transformers*

- Engineered a secure, multi-user banking chatbot with JWT role-based authentication (Admin/User), protecting 10+ REST API endpoints (/chat, /balance, /statement, /profile); designed full MySQL schema and an HTML/CSS/JS frontend exchanging data via JSON REST APIs.
- Implemented Retrieval-Augmented Generation (RAG) using sentence-transformer embeddings and cosine similarity-based search, enabling AI-powered FAQ retrieval from a structured knowledge base with contextually accurate responses.

SQL Business Analytics Practice | *MySQL · Advanced Query Engineering*

- Solved 200+ real-world SQL business problems on employee, department, payroll, and organizational datasets applying joins, self-joins, correlated subqueries, aggregate functions, date/string functions, and rank-based filtering logic.
- Performed department-wise reporting, salary band analysis, employee-manager hierarchy mapping, hiring trend analysis, and business KPI calculations, directly reinforcing skills evaluated in Data Analyst and Business Analyst technical interviews.

DataMentor – No-Code Machine Learning Platform | *Python · Flask · LLM Integration*

- Developed a web-based no-code supervised ML platform enabling non-technical users to upload datasets, run automated EDA, handle missing values and outliers, train classification and regression models, and generate business-aligned interpretation reports.
- Integrated an LLM-powered assistant to translate dataset behavior, model logic, and results into plain-language explanations, bridging technical ML workflows with business stakeholders and enhancing decision support.

PROOF – Deepfake Detection & Media Integrity System | *Python · LBP · Gabor Filters · OpenCV*

- Built a modular deepfake detection pipeline using Local Binary Patterns (LBP), Gabor filters, and facial landmark analysis to detect video-level manipulation anomalies in eyes, iris, and skin texture with targeted specificity.
- Developed an interpretable model architecture that pinpoints specific manipulation cues, improving robustness against high-quality synthetic media and supporting forensic media authenticity analysis.

LEADERSHIP & EXTRACURRICULAR

Treasurer, IEEE Student Chapter | *Christ University, Bengaluru | Jan 2025 – Apr 2026*

- Managed the chapter treasury and maintained accurate financial records across 2 flagship technical events; coordinated and split the allocated budget across a 12-member core committee covering event logistics, design, and operations while ensuring all activities stayed within approved limits.
- Led end-to-end event execution from the ground up — including poster design, team coordination, and on-ground management — consolidating inputs from all 12 committee members and translating collective ideas into structured, well-delivered campus events.
- Planned and coordinated budgets for technical events, workshops, and seminars across three academic semesters, maintaining transparent financial documentation throughout each event cycle.
- Collaborated with the IEEE core committee to organise and execute technical activities on campus, contributing to student professional development and fostering a culture of innovation and technology engagement.

CERTIFICATIONS

- AWS Certified Cloud Practitioner – Udemy
- Full Stack Web Development – Udemy
- Python Programming – CSC